Unit: Unit 0: Foundations of Algebra
Topic: Rational Numbers Decimals

Name: __________
Date: __________

Homework (Habanero)

Q1. What is the decimal equivalent of the fraction $\frac{3}{8}$?
(A) 0.35
(B) 0.375
(C) 0.4
(D) 0.42
(E) 0.45

Q2. A patient is prescribed $2\frac{1}{4}$ milligrams of medication. What is this dosage in decimal form?
(A) 2.1
(B) 2.25
(C) 2.3
(D) 2.4
(E) 2.5

Q3. What percent of 240 is 48?
(A) 10
(B) 15
(C) 20
(D) 25
(E) 30

Q4. Order the rational numbers -0.5, $\frac{1}{4}$, and 0.2 from least to greatest.
(A) -0.5, 0.2, $\frac{1}{4}$
(B) -0.5, $\frac{1}{4}$, 0.2
(C) 0.2, $\frac{1}{4}$, -0.5
(D) $\frac{1}{4}$, 0.2, -0.5
(E) 0.2, -0.5, $\frac{1}{4}$

Q5. A water tank can hold 240 gallons of water. If $\frac{3}{8}$ of the tank is already filled, how many more gallons can be added?
(A) 180
(B) 160
(C) 144
(D) 120
(E) 100

Q6. What is the sum of 2.5 and $1\frac{1}{4}$?
(A) 3.5
(B) 3.7
(C) 3.9
(D) 4.1
(E) 4.3

Q7. A cell divides into $\frac{3}{4}$ of its original size every hour. If the cell is initially 10 millimeters in diameter, what will its diameter be after 2 hours?
(A) 4.5
(B) 5.625
(C) 6.25
(D) 7.5
(E) 7.8125

Q8. A medication has a 25% chance of being effective. If 16 patients are given the medication, what is the expected number of patients for whom the medication will be effective?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Open Ended Questions (FRQ)

Q9. A hospital is conducting a study on patient recovery times. The average recovery time for a certain procedure is 5.2 days, with a standard deviation of 1.1 days. If a patient takes 7.5 days to recover, how many standard deviations away from the mean is this patient's recovery time?

Q10. The population of a certain type of cell grows according to the equation $P(t) = 200(1.05)^t$, where t is time in hours. If the population is currently 400 cells, how many hours will it take for the population to double?

Chef's Tips

- Emphasize the importance of significant figures in calculations.
- Remind students to label units in their answers.
- Encourage students to check their work by plugging their answers back into the original equations.